

# Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Red-headed Woodpecker"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

**Check whether MCMC worked:**

**Do we have divergent transitions?**

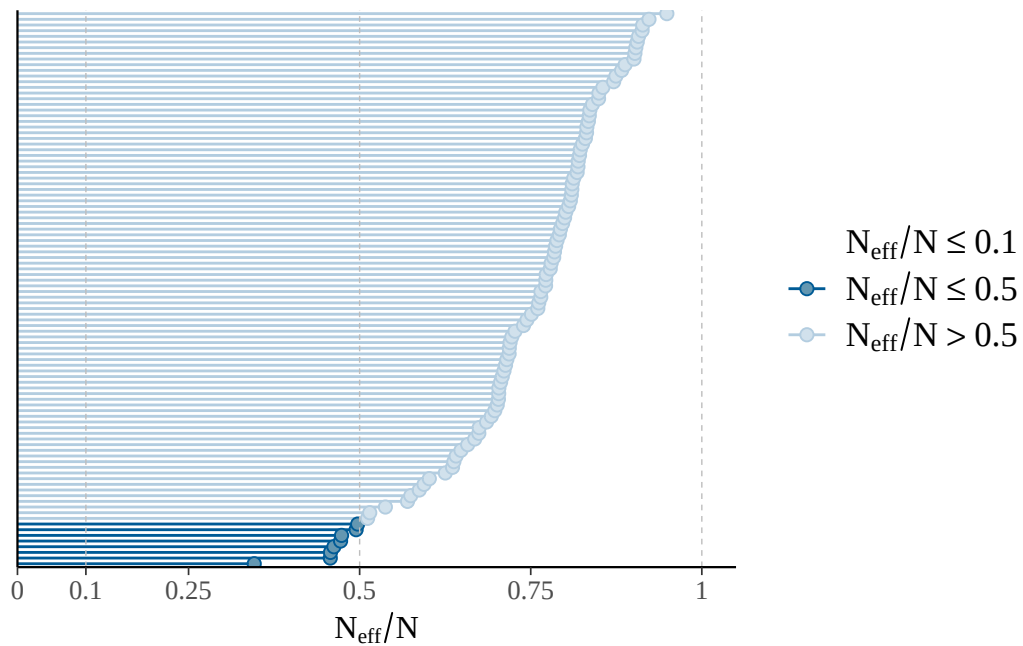
```
[1] "divergent transitions: 0"
```

**Explorations regarding divergent transitions:**

...

**Is effective number of MCMC samples large enough:**

The larger the ratio of  $n_{\text{eff}}$  to  $N$  (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.4574 | 0.9798  | 1.1836 | 1.1357 | 1.3387  | 2.0831 |

```
[1] "n_eff ratio tail ESS:"
```

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.4571 | 0.7001  | 0.7718 | 0.7541 | 0.8231  | 0.9228 |

```
[1] "effective sample size in bulk and tail fine"
```

**Do chains mix well:**

No divergences to plot.

No divergences to plot.



No divergences to plot.





Post-warmup iteration

### Are R-hat values fine:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.9991 | 0.9995  | 0.9997 | 0.9999 | 1.0002  | 1.0012 |

```
[1] "All R-hat values < 1.01"
```

### Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_bio1_3yrs" "b_occ_pr_winter_3yrs"
```

```
[3] "b_occ_lpr_winter_3yrsE2" "b_occ_sum_annual_crops_3yrs"
```

```
[1] "b_colo_bio1" "b_colo_pr_winter"
```

```
[3] "b_colo_sum_annual_crops" "b_colo_secdf"
```

```
[5] "b_colo_lsecdfE2"
```

```
[1] "b_ex_bio1" "b_ex_pr_summer" "b_ex_lbio1E2"
```

```
[4] "b_ex_sum_annual_crops" "b_ex_pastr" "b_ex_lsecdfE2"
```

```
[7] "b_ex_lurbanE2"
```

